

YOLOv8

requirements.txt

```
# Object Detection Course
```

```
cvzone==1.5.6  
ultralytics==8.0.26  
hydra-core>=1.2.0  
matplotlib>=3.2.2  
numpy>=1.18.5  
opencv-python==4.5.4.60  
Pillow>=7.1.2  
PyYAML>=5.3.1  
requests>=2.23.0  
scipy>=1.4.1  
torch>=1.7.0  
torchvision>=0.8.1  
tqdm>=4.64.0  
filterpy==1.4.5  
scikit-image==0.19.3  
lap==0.4.0
```

```
pip install -r requirements.txt
```

YOLOv8 첫번째 프로그램 실행하기

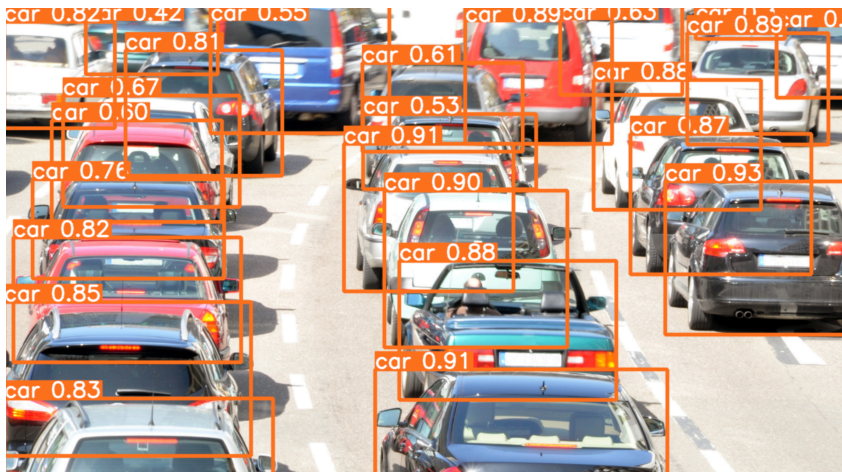
```
from ultralytics import YOLO
import cv2

model = YOLO("yolov8n.pt")

results = model("Images/1.png", show=True)
cv2.waitKey(0)
```

이미지 무료 사이트에서 car, motorcycle, person 이미지 다운로드
<https://unsplash.com/ko>

YOLOv8 첫번째 프로그램 실행하기



YOLOv8 첫번째 프로그램 실행하기

```
model = YOLO("../Yolo-Weights/yolov8l.pt")
```

YOLO-Webcam

```
from ultralytics import YOLO
import cv2
import cvzone
import math
import time

cap = cv2.VideoCapture(cv2.CAP_DSHOW + 0)
cap.set(cv2.CAP_PROP_FRAME_WIDTH, 640)
cap.set(cv2.CAP_PROP_FRAME_HEIGHT, 480)

model = YOLO("../Yolo-Weights/yolov8l.pt")

while True:
    success, img = cap.read()
    results = model(img, stream=True)

    cv2.imshow("Image", img)
    cv2.waitKey(1)
```

```
while True:
    success, img = cap.read()
    results = model(img, stream=True)

    for r in results:
        boxes = r.boxes

        print(len(boxes))
        for box in boxes:
            x1, y1, x2, y2 = box.xyxy[0]
            print("tensor x1, y1, x2, y2 ", x1, y1, x2, y2)
            x1, y1, x2, y2 = int(x1), int(y1), int(x2), int(y2)
            print("x1, y1, x2, y2 ", x1, y1, x2, y2)

            cv2.rectangle(img, (x1,y1),(x2,y2), (255,0,255), 3)

cv2.imshow("Image", img)
cv2.waitKey(1)
```



```
while True:
    success, img = cap.read()
    results = model(img, stream=True)
```

```
for r in results:
    boxes = r.boxes
    for box in boxes:
        # Bounding Box
        x1, y1, x2, y2 = box.xyxy[0]
        x1, y1, x2, y2 = int(x1), int(y1), int(x2), int(y2)
        w, h = x2 - x1, y2 - y1
        cvzone.cornerRect(img, (x1, y1, w, h))
        conf = math.ceil((box.conf[0] * 100)) / 100 # Confidence
```



```
        cvzone.putTextRect(
            img,
            f"{conf}",
            (max(0, x1), max(35, y1)),
            scale=1,
            thickness=1,
        )
```



```
cv2.imshow("Image", img)
cv2.waitKey(1)
```

YOLO-Webcam

```
from ultralytics import YOLO
import cv2
import cvzone
import math
import time

cap = cv2.VideoCapture(cv2.CAP_DSHOW + 0)
cap.set(cv2.CAP_PROP_FRAME_WIDTH, 1280)
cap.set(cv2.CAP_PROP_FRAME_HEIGHT, 720)

model = YOLO("../Yolo-Weights/yolov8l.pt")

# 80개
classNames = [
    "person",
    "bicycle",
    "parking meter",
    "bench",
]
```

```
while True:

    success, img = cap.read()
    results = model(img, stream=True)
    for r in results:
        boxes = r.boxes
        for box in boxes:
            # Bounding Box
            x1, y1, x2, y2 = box.xyxy[0]
            x1, y1, x2, y2 = int(x1), int(y1), int(x2), int(y2)

            w, h = x2 - x1, y2 - y1
            cvzone.cornerRect(img, (x1, y1, w, h))
            conf = math.ceil((box.conf[0] * 100)) / 100 # Confidence

            cls = int(box.cls[0]) # Class Name

            cvzone.putTextRect(
                img,
                f"{classNames[cls]} {conf}",
                (max(0, x1), max(35, y1)),
                scale=1,
                thickness=1,
            )

    cv2.imshow("Image", img)
    cv2.waitKey(1)
```